Given these sample points:

[0.17152521 0.34391687 0.2487372 ]
[0.24211446 0.64407427 0.27243281]
[0.53490572 0.39850092 0.17338873]
[0.49258141 0.61159319 0.34017639]
[0.13462167 0.21991724 0.45820622]
[0.34552327 0.94135983 0.26936348]
[0.15183663 0.43999062 0.99088187]
[0.64550284 0.39714294 0.91977134]
[0.74691195 0.28419631 0.22629985]
[0.17047699 0.6970324  0.14916943]
[0.22054934 0.29782524 0.34355534]
[0.66601366 0.67198515 0.2462953 ]
[0.04680895 0.23136024 0.77061759]
[0.60009728 0.72513573 0.06608864]
[0.96599485 0.86111969 0.56682913]
[0.552154 0.001705 0.636366]
[0.812901 0.999718 0.996637]
[0.98663  0.965119 0.00494 ]
[0.997382 0.392157 0.7232  ]
[0.998198 0.003505 0.02763 ]
[4.09189e-01 9.91786e-01 8.54000e-04]
[0.401619 0.413251 0.491273]
[0.335009 0.207283 0.377797]

And these results:

-0.11212220046256897
-0.08796286022736445
-0.111414654295324
-0.034835313350078584
-0.04800758439218157
-0.11062091307282658
-0.3989255131463011
-0.11386851478863991
-0.13146060864136055
-0.09418956091057398
-0.04694740582651916
-0.10596503573558178
-0.11804825644688696
-0.036377828071632486
-0.05675837155397648
-0.16554079187169876
-0.46825413015901896
-0.11861332481955808
-0.13535893134164836
-0.11866469653270655
-0.14049698989772344
-0.021075704193875416
-0.04263921555802391

A small positive number is bigger than a large negative number

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially doing **black-box optimization** over a 3D space $[0, 1]^3$, trying to **maximize** a function given noisy samples — and all your current observations are negative, so you're looking for the *least negative (i.e., closest to zero or positive)* region.

### Key observations from your data

- Best (highest) value so far:
  **−0.0211 at** `[0.4016, 0.4133, 0.4913]`
- Next best values cluster around:
  - mid-range $x \approx 0.3$–$0.5$
  - mid-range $y \approx 0.2$–$0.5$
  - mid-range $z \approx 0.3$–$0.5$

### Patterns

- Very high values of any coordinate (near 1.0) tend to perform worse.
- Very low values (near 0.0) also tend to perform worse.
- The **central region** looks promising.
- There's **no sharp optimum yet**, so exploration near the best point is ideal.

---

### Recommended next sampling strategy

Since you want:

- **Temperature = 0.1** → mostly exploit (stay near best)
- **Top-k = 3, Top-p = 3** → choose among top candidates, low diversity

We should generate a few candidates near the current best and pick one.

### Top candidate points (ranked)

**1. Best next sample (exploit slightly)**

```csharp
[0.42, 0.45, 0.50]
```

**2. Slight variation (local exploration)**

```csharp
[0.38, 0.40, 0.52]
```

**3. Another nearby probe**

```csharp
[0.45, 0.38, 0.47]
```

---

### Why this works

This mimics **Bayesian optimization intuition**:

- You already found a promising basin near `[0.40, 0.41, 0.49]`
- With low temperature, you:
  - **refine locally** instead of jumping randomly
- These points:
  - Stay close enough to exploit
  - Add slight perturbations to detect gradient direction

---

### If you want a single answer:

👉 **Sample next at:**

```csharp
[0.42, 0.45, 0.50]
```

---

### Optional next step

If this improves:

- Keep doing **small Gaussian steps** around the best point

If it worsens:

- Expand radius slightly (e.g., ±0.1)

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If you want, I can:

- Fit a surrogate model (Gaussian Process style)
- Or generate a full acquisition-function-based next point (EI/UCB)